Angles of polygons

1 360°

2 360°

3

a $y=63^{\circ}$

b $x=64^{\circ}$

4

a 4 triangles

b 720°

5

a 4 triangles

b 720°

6

Name of the polygon	Number of sides	Least number of triangles	Interior angle sum
Pentagon	5	3	540°
Hexagon	6	4	720°
Heptagon	7	5	900°
Octagon	8	6	1080°
Nonagon	9	7	1260°
Decagon	10	8	1440°

7

a

i 180° ii 180° iii 180°

b 540°

c $n-2$

d $(n-2)\times 180$

8 1260°

9

a

i $x=73^{\circ}$

ii $a=88^{\circ}$

iii $b=92^{\circ}$

iv $c=107^{\circ}$

v $d=73^{\circ}$

b 360°

10

a $y = 158^\circ$

b $x = 81^\circ$

c $x = 139^\circ, y = 101^\circ$

11

a $x = 108^\circ$

b $x = 120^\circ$

12 $x = 7^\circ$

13 $x = \dfrac{(n-2) \times 180}{n}$

14

a 120°

b 135°

c 144°

d 150°

15 18°

16

a $x = 135$

b $y = 135$

17 $x = 72^\circ$

18

a $x = 177^\circ$

b $y = 98$

19 $x = \dfrac{360}{n}$

20

a

i 12

ii Dodecagon

b

i 9

ii Nonagon

c

i 6

ii Hexagon

d

i 10

ii Decagon

e

i 5

ii Pentagon

i 8

ii Octagon

g

i 4.8

ii This shape cannot exist.

h

i 7.2

ii This shape cannot exist.

i

i 4.5

ii This shape cannot exist.

j

i 7.2

ii This shape cannot exist.

Additional questions

21

a 60°

b 135°

c 51.43°

d 16.36°

22

a 1440°

b 1800°

c 540°

d 3600°

23 360°

24

a $x = 74$

b $x = 96$

25

a $x = 91$

b $x = 25$

26

a $m\angle A = 30^\circ$

b $m\angle FGH = 122^\circ$

c $m\angle HIJ = 116^\circ$

d 360°

27 $x = 104$

28

a 4 sides

b 8 sides

c 15 sides

d 24 sides

e 13 sides

f 18 sides

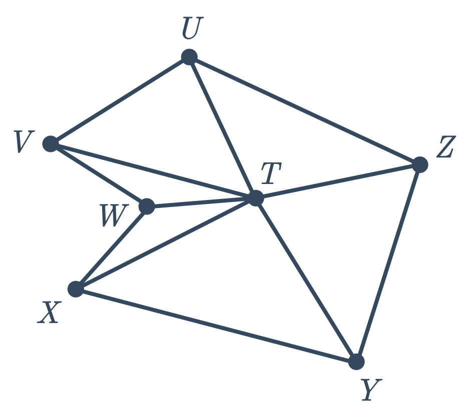
29

a $m\angle AED = 108^\circ$. We know that the sum of the interior angles of a regular polygon is 540°. Since all interior angles in a regular polygon are congruent, we can perform $540^\circ \div 5 = 108^\circ$ to get the measure of each interior angle.

b We know each interior angle is 108°. In $\triangle AGH$, $\overline{AG} \cong \overline{AH}$ so $m\angle AGH = m\angle AHG = 72^\circ$. Using triangle sum theorem, $m\angle GAH = 36^\circ$. Using angle addition, we get $m\angle EAG + m\angle HAB + 36^\circ = 108^\circ$. We can show with congruent triangles that $\angle EAG \cong \angle HAB$. so we can use substitution to get $x + x + 36^\circ = 108^\circ$. Solving the equation for x gets the result $x = 36^\circ$.

30

a



b Using triangle sum theorem we get:

$m\angle VTW + m\angle TVW + m\angle TWV = 180^\circ$
$m\angle VTU + m\angle TUV + m\angle TVU = 180^\circ$
$m\angle UTZ + m\angle TUZ + m\angle TZU = 180^\circ$
$m\angle YTZ + m\angle TYZ + m\angle TZY = 180^\circ$
$m\angle YTX + m\angle TYX + m\angle TXY = 180^\circ$
$m\angle XTW + m\angle TXW + m\angle TWX = 180^\circ$
And $6 \times 180^\circ = 1080^\circ$

c The six angles that meet at T form a complete circle, meaning that $m\angle WTV + m\angle VTU + m\angle UTZ + m\angle ZTY + m\angle YTX + m\angle XTW = 360^\circ$

d From part a) we get that the sum of all of the angles in the six triangles is 1080°. If we remove the six angles that meet at T we are left with the interior angles of Hexagon $UVWXYZ$. So, we can perform subtraction using the result from part b):
$1080^\circ - (m\angle WTV + m\angle VTU + m\angle UTZ + m\angle ZTY + m\angle YTX + m\angle XTW) =$
$1080^\circ - 360^\circ = 720^\circ$

31 The interior angles of a reguar pentagon each have a measure of 108°. The interior angles of a regular hexagon each have a measure of 120°. If you look at the vertex where a pentagon and a hexagon meet, the third angle at this point is equal to:

$360^\circ - 108^\circ - 120^\circ = 132^\circ$,

meaning that neither a reguar hexagon or pentagon could connect at the vertex without any gaps.